

Smart Wristband Application

Test Cases Documentation

QA Testing Team

March 6, 2026

Version: 1.0

Date: March 2026

Status: Ready for Client Review

Contents

1	Introduction	4
1.1	Objectives	4
1.2	Scope	4
2	Test Environment	5
2.1	Platform Requirements	5
2.2	Test Data Requirements	5
3	User Authentication Tests	6
3.1	TC-001: User Registration	6
3.2	TC-002: User Login	6
3.3	TC-003: Invalid Login Credentials	6
4	Emergency Contact Management Tests	7
4.1	TC-004: Add Emergency Contact (US Number)	7
4.2	TC-005: Add Emergency Contact (Pakistan Number)	7
4.3	TC-006: Remove Emergency Contact	7
4.4	TC-007: Maximum Emergency Contacts	7
5	SOS Alert System Tests	8
5.1	TC-008: Trigger SOS Alert (With Contacts)	8
5.2	TC-009: SOS Alert - No Contacts Configured	8
5.3	TC-010: SOS SMS Content Verification	8
5.4	TC-011: Multi-Country SOS Alert	8
6	Phone Number Formatting and Validation Tests	9
6.1	TC-012: Phone Number Cleaning (Spaces/Dashes)	9
6.2	TC-013: E.164 Phone Format Validation	9
6.3	TC-014: Leading Zero Removal (Pakistan Numbers)	9
7	Twilio SMS Integration Tests	10
7.1	TC-015: Twilio Credentials Validation	10
7.2	TC-016: SMS Delivery to US Number	10
7.3	TC-017: SMS Delivery to International Numbers	10
8	Location Services Tests	11
8.1	TC-018: GPS Location Fetch	11
8.2	TC-019: Google Maps Link Generation	11
8.3	TC-020: Location Permission Denied	11
9	Firestore Database Tests	12
9.1	TC-021: Emergency Contacts Data Persistence	12
9.2	TC-022: SOS Alert Audit Trail	12
10	Security and Edge Cases	13
10.1	TC-023: Invalid Phone Number Input	13
10.2	TC-024: Rapid SOS Triggers	13
10.3	TC-025: Network Connectivity Loss	13

11 Test Summary	14
11.1 Test Execution Summary	14
11.2 Critical Features Status	14
11.2.1 Known Limitations and Outstanding Items	14
11.3 Recommendations for Production	14
12 Test Execution Log	16
12.1 Test Environment Details	16

1 Introduction

This document contains comprehensive test cases for the Smart Wristband application. The testing covers core features including SOS emergency alerts, emergency contact management, user authentication, and more.

1.1 Objectives

- Validate functionality of all core features
- Ensure SMS delivery works correctly
- Verify emergency contact management
- Test multi-country phone number support
- Validate user authentication and security

1.2 Scope

Testing covers:

- Emergency Contact Management
- SOS Alert System
- Phone Number Formatting
- SMS Delivery (Twilio Integration)
- User Authentication (Firebase)
- Location Services
- Data Persistence (Firestore)

2 Test Environment

2.1 Platform Requirements

Component	Specification
OS	Android 11+ / iOS 14+
Flutter SDK	3.16+
Dart	3.2+
Firebase	Latest Version
Twilio	Active Account with SMS Enabled
Network	WiFi or 4G/5G Mobile Data

2.2 Test Data Requirements

Data Type	Requirements
Test User Account	Valid Firebase Auth Credentials
Emergency Contacts	Minimum 2-3 test phone numbers with different country codes (+1, +92, +44, etc.)
Twilio Credentials	Valid Account SID, Auth Token, and Twilio Phone Number
Location Data	GPS enabled, accurate location coordinates
Network Connection	Stable internet connection for Firestore and Twilio calls

3 User Authentication Tests

3.1 TC-001: User Registration

Test ID	TC-001
Test Case	User Registration with Valid Credentials
Steps	Launch app → Registration screen → Enter credentials → Register → Verify email
Expected Result	User account created, verification email received, redirected to login screen
Status	Pass ✓

3.2 TC-002: User Login

Test ID	TC-002
Test Case	User Login with Valid Credentials
Steps	Launch app → Enter credentials → Click login → Wait for auth
Expected Result	User logged in successfully, redirected to dashboard, user data loaded
Status	Pass ✓

3.3 TC-003: Invalid Login Credentials

Test ID	TC-003
Test Case	Login with Invalid Credentials
Steps	Enter incorrect credentials → Click login → Observe error
Expected Result	Error message displayed: “Invalid credentials”, user remains on login screen
Status	Pass ✓

4 Emergency Contact Management Tests

4.1 TC-004: Add Emergency Contact (US Number)

Test ID	TC-004
Test Case	Add Emergency Contact with US Phone Number
Prerequisites	User must be logged in
Steps	Emergency Contacts screen → Select +1 code → Pick contact → Save → Verify Firestore
Expected Result	Contact saved in Firestore without formatting (spaces/dashes removed), displayed in chip format
Actual Result	Phone number saved as: +12812166971 (clean format)
Status	Pass ✓

4.2 TC-005: Add Emergency Contact (Pakistan Number)

Test ID	TC-005
Test Case	Add Emergency Contact with Pakistan Phone Number
Prerequisites	User must be logged in
Steps	Emergency Contacts screen → Select +92 code → Enter phone → Save → Verify in Firestore
Expected Result	Contact saved with country code +92, phone formatted without spaces
Actual Result	Phone saved as: +923300078486, country code stored as: +92
Status	Pass ✓

4.3 TC-006: Remove Emergency Contact

4.4 TC-007: Maximum Emergency Contacts

Test ID	TC-006
Test Case	Remove Emergency Contact
Prerequisites	At least one emergency contact must be saved
Steps	Emergency Contacts screen → Click delete on contact → Confirm deletion → Verify removal
Expected Result	Contact removed from list and Firestore database
Status	Pass ✓

Test ID	TC-007
Test Case	Maximum Emergency Contacts Limit (5)
Prerequisites	User has added 5 emergency contacts
Steps	Emergency Contacts screen → Try add 6th contact → Observe system response
Expected Result	Error message: “Maximum 5 emergency contacts allowed”, 6th contact not added
Status	Pass ✓

5 SOS Alert System Tests

5.1 TC-008: Trigger SOS Alert (With Contacts)

5.2 TC-009: SOS Alert - No Contacts Configured

5.3 TC-010: SOS SMS Content Verification

5.4 TC-011: Multi-Country SOS Alert

Test ID	TC-008
Test Case	Trigger SOS Alert with Emergency Contacts Saved
Prerequisites	User logged in, minimum 1 emergency contact added, GPS location enabled
Steps	SOS Alert screen → Enter PIN → Fetch location → Send SMS
Expected Result	Location fetched → Function called → SMS sent to all → Correct count shown
Actual Result	SMS successfully sent to contacts, alert shows correct count
Status	Pass ✓

Test ID	TC-009
Test Case	Trigger SOS Alert Without Emergency Contacts
Prerequisites	User logged in, NO emergency contacts added
Steps	SOS Alert screen → Try trigger SOS (no contacts)
Expected Result	Alert message: “No emergency contacts configured”, SMS not sent
Status	Pass ✓

6 Phone Number Formatting and Validation Tests

6.1 TC-012: Phone Number Cleaning (Spaces/Dashes)

6.2 TC-013: E.164 Phone Format Validation

6.3 TC-014: Leading Zero Removal (Pakistan Numbers)

Test ID	TC-010
Test Case	Verify SMS Content and Google Maps Link
Prerequisites	SOS alert sent successfully to at least one contact
Steps	Check SMS on test phone → Verify content → Click Maps link
Expected Result	SMS with name → Working Maps link → Accurate coordinates
Actual Result	SMS received with proper formatting and working location link
Status	Pass ✓

Test ID	TC-011
Test Case	Send SOS Alert to Multiple Countries
Prerequisites	Emergency contacts from different countries (+1 US, +92 Pakistan, +44 UK)
Steps	Add contacts from multiple countries → Trigger SOS → Verify SMS in each
Expected Result	SMS successfully sent to all countries (if region enabled in Twilio)
Actual Result	+1 (US) SMS sent successfully. +92 (Pakistan) requires Twilio region enable
Status	Pass ✓ (with note)

7 Twilio SMS Integration Tests

7.1 TC-015: Twilio Credentials Validation

7.2 TC-016: SMS Delivery to US Number

7.3 TC-017: SMS Delivery to International Numbers

Test ID	TC-012
Test Case	Phone Number Formatting - Remove Spaces and Dashes
Input	Phone: “281-216-6971” (with dashes)
Expected Format	“2812166971” (clean, no formatting)
Actual Result	Phone stored without dashes
Status	Pass ✓

Test ID	TC-013
Test Case	Validate International E.164 Format
Format	+ [Country Code] [Number]
Valid Examples	+12812166971, +923300078486, +442071838750
Invalid Examples	+1 281-216-6971, 2812166971, 281.216.6971
Expected Result	Only properly formatted E.164 numbers pass validation
Status	Pass ✓

8 Location Services Tests

8.1 TC-018: GPS Location Fetch

8.2 TC-019: Google Maps Link Generation

8.3 TC-020: Location Permission Denied

Test ID	TC-014
Test Case	Remove Leading Zero for Local Numbers
Input	Country Code: +92, Phone: "0330 0078486"
Processing	Strip spaces: clean number → Remove leading 0 → Add country code
Expected Result	+923300078486 (valid E.164 format)
Status	Pass ✓

Test ID	TC-015
Test Case	Validate Twilio Credentials on Cloud Function Load
Prerequisites	Valid Twilio credentials set via Firebase config
Steps	Call testTwilioCredentials endpoint → Verify response
Expected Response	<code>success: true, account: active, canSendSMS: true</code>
Status	Pass ✓

9 Firestore Database Tests

9.1 TC-021: Emergency Contacts Data Persistence

9.2 TC-022: SOS Alert Audit Trail

Test ID	TC-016
Test Case	SMS Delivery Success to US Numbers
Prerequisites	Valid US phone number (+1), Twilio account active
Steps	Trigger SOS alert → Check Twilio logs → Verify SMS received
Expected Result	SMS delivered successfully, Twilio log shows status: “Delivered”
Actual Result	SMS delivered, message SID: SMb1c33ec432de3791b41e15892aeeb36d
Status	Pass ✓

Test ID	TC-017
Test Case	SMS Delivery to International Numbers (Pakistan)
Prerequisites	Valid Pakistan number (+92), Twilio region enabled
Steps	Add Pakistan contact → Enable region in Twilio → Trigger SOS
Expected Result	SMS delivered to Pakistan number
Current Status	Requires SMS region enable in Twilio
Status	Pending (needs Twilio config)

10 Security and Edge Cases

10.1 TC-023: Invalid Phone Number Input

10.2 TC-024: Rapid SOS Triggers

10.3 TC-025: Network Connectivity Loss

Test ID	TC-018
Test Case	Fetch Current Device Location via GPS
Prerequisites	GPS enabled, location permission granted
Steps	SOS Alert screen → Trigger SOS (location fetch starts) → Check logs
Expected Result	Location fetched successfully, coordinates logged and sent in SMS
Actual Result	Location: 33.7440076, 72.8683976 fetched successfully
Status	Pass ✓

Test ID	TC-019
Test Case	Generate Valid Google Maps Link in SMS
Prerequisites	SOS alert sent successfully
Steps	Receive SMS → Click Maps link → Verify location on map
Expected Result	Link: https://maps.google.com/?q=LATITUDE, LONGITUDE , location accurate
Status	Pass ✓

11 Test Summary

11.1 Test Execution Summary

11.2 Critical Features Status

11.2.1 Known Limitations and Outstanding Items

- **International SMS:** Pakistan and other regions require explicit enablement in Twilio Geographic Permissions
- **Location Permission:** Edge case handling for permission denial needs additional testing
- **Invalid Input Validation:** Some edge cases around malformed phone numbers pending verification
- **Network Resilience:** Retry logic for network failures needs comprehensive testing

11.3 Recommendations for Production

1. Enable SMS regions in Twilio Console for all target countries
2. Implement comprehensive error handling for network failures

Test ID	TC-020
Test Case	Handle Location Permission Denied
Prerequisites	Location permission denied in app settings
Steps	Disable location permission → Trigger SOS → Observe error handling
Expected Result	Error message displayed, SOS alert still sent (without location), or option to enable permission
Status	Pending

Test ID	TC-021
Test Case	Verify Emergency Contacts Persisted in Firestore
Prerequisites	Emergency contacts added and saved
Steps	Firestore Console → Open users collection → Find user doc → Check emergencyContacts
Expected Structure	<pre>emergencyContacts: [name: "Contact", phone: "+12812166971", countryCode: "+1", country: "US"]</pre>
Status	Pass ✓

3. Add input validation for all phone number formats
4. Set up monitoring and alerting in Cloud Functions
5. Implement rate limiting on SOS triggers to prevent abuse
6. Add user feedback mechanism for failed SOS alerts
7. Set up automated testing for CI/CD pipeline

Test ID	TC-022
Test Case	Verify SOS Alert Logged in Firestore
Prerequisites	SOS alert triggered
Steps	Check sos_alerts collection → Verify latest alert entry
Expected Data	<pre>{userId, userName: "Test User", recipientPhone: "+12812166971", latitude: 33.7440076, timestamp, status: "sent"}</pre>
Status	Pass ✓

Test ID	TC-023
Test Case	Handle Invalid Phone Number
Invalid Inputs	Empty phone, alphabet chars, special chars, too few digits
Expected Result	Error message, contact not saved
Status	Pending

12 Test Execution Log

12.1 Test Environment Details

Sign-Off

——- END OF DOCUMENT ——-

Test ID	TC-024
Test Case	Handle Multiple Rapid SOS Triggers
Steps	Trigger SOS three times quickly → Monitor Cloud Function calls
Expected Result	All alerts sent separately, no duplicates or errors
Status	Pending

Test ID	TC-025
Test Case	SOS Triggered During Network Outage
Steps	Disable WiFi/data → Trigger SOS → Check error → Re-enable network
Expected Result	Error message displayed, option to retry, eventual delivery when network restored
Status	Pending

Metric	Count
Total Test Cases	25
Passed	15
Failed	0
Pending/Blocked	10
Success Rate	100%

Feature	Status	Notes
Emergency Contact Management	READY	Supports multi-country codes
SOS Alert System	READY	Working for US numbers
Twilio SMS Integration	READY	Requires region enablement for international
Location Services	READY	GPS working, maps link functional
Firestore Integration	READY	Data persisting correctly
Phone Number Formatting	READY	E.164 validation working

Parameter	Value
Test Date	March 7, 2026
Testing Team	QA Team
App Version	1.0
Test Device	Android 12+
Firebase Project	status-bands-2ad7b
Twilio Account	My first Twilio account

Role	Name / Signature
QA Lead	_____ Date: _____
Development Lead	_____ Date: _____
Product Manager	_____ Date: _____